Method of Distribution Functions

X, Y denote r.v.

x: – takes value and outputs another

Note: We will only consider continuous random variables for this method.

$x_i = 1, 2, 3, 4, 5, 6$

$P(x_i) = \frac{1}{6}$

Consider $U = h(Y_1, \ldots, Y_n)$ to be a function of the continuous random variables $Y_1, \ldots, Y_n$, which have joint density $f(y_1, \ldots, y_n)$.

$P(X \le x) = \int_{-\infty}^{x} f_X(x)\, dx$

- Find the region in the $(y_1, \ldots, y_n)$ space where $U \le u$.

- Find $F_U(u) = P(U \le u) = P(h(Y_1, \ldots, Y_n) \le u) = \int \cdots \int_{U \le u} f(y_1, \ldots, y_n) dy_1 \cdots dy_n$.

- Then, the density of U is $f_U(u) = \frac{d}{du} F_U(u)$.

The amount of time waiting Y,

Density function of Y : $f(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta}, & y > 0 \\ 0, & \text{otherwise} \end{cases}$

Let $U = aY + b$, $a > 0$.

Distribution of U $F_U(u)$

$$F_U(u) = P(U < u) = P(\underbrace{aY + b < u})$$

$$P\left(Y < \frac{u-b}{a}\right) = \int_{-\infty}^{\frac{u-b}{a}} f_Y(y)\, dy$$

2 cases, $\frac{u-b}{a} < 0$, $u < b$ $\therefore \int_{-\infty}^{\frac{u-b}{a}} f_Y(y)\, dy = 0$

or $\frac{u-b}{a} \geq 0$, $u \geq b$ $\therefore \int_{-\infty}^{\frac{u-b}{a}} f_Y(y)\, dy = \int_{0}^{\frac{u-b}{a}} f_Y(y)\, dy = \underbrace{\int_{0}^{\frac{u-b}{a}} \frac{1}{\theta} e^{\frac{-y}{\theta}}\, dy}_{\text{Calculate and find}}$

$$\int_{0}^{\frac{u-b}{a}} \frac{1}{\theta} e^{-y/\theta}\, dy$$

$$= -e^{-y/\theta} \Big|_{0}^{\frac{u-b}{a}} = -e^{\frac{-u+b}{a\theta}} - (-e^{0})$$

$$f_U(u) = \frac{1}{a\theta} e^{\frac{-u+b}{a\theta}}, \quad (u > b)$$

$$f_U(u) = \begin{cases} \frac{1}{a\theta} e^{-\frac{u+b}{a\theta}} & , u > b \\ 0 & , \text{otherwise} \end{cases}$$

Q. $U = Y^a \quad (a > 0)$

$$F_U(u) = P(U < u) = P(Y^a < u)$$

$$P(Y < \sqrt[a]{U}) = \int_{-\infty}^{U^{\frac{1}{a}}} f_Y(y)\, dy$$, we want $U^{1/a} > 0 \rightarrow U > 0$

in the case $U > 0$

$$\int_0^{u^{1/a}} \frac{1}{\theta} e^{-y/\theta}\, dy = -e^{-y/\theta}\Big|_0^{u^{1/a}} = -e^{\frac{-u^{1/a}}{\theta}} - (-e^{-0})$$

$$= 1 - e^{\frac{-u^{1/a}}{\theta}}, \quad u > 0.$$

Otherwise $f_Y(y) = 0$

$$F_U(u) = \frac{d}{dx}\left(1 - e^{\frac{-u^{1/a}}{\theta}}\right) = -e^{-\frac{u^{1/a}}{\theta}}\left(-\frac{1}{\theta}\cdot\frac{1}{a}u^{(1/a-1)}\right)$$

$$= \frac{1}{\theta a}\cdot u^{(1/a-1)} e^{\frac{-u^{1/a}}{\theta}}, \quad u > 0$$

0, otherwise

Y is a r.v.

dist. function

$$f_Y(y) = \begin{cases} \frac{3y^2}{16} & -2 < y < 2 \\ 0, & \text{oth.} \end{cases}$$

Density function

$U = 1 - 6Y^2$

$$F_U(u) = P(U \le u) = P(1 - 6Y^2 \le u) = P\left(\frac{1-u}{6} \le Y^2\right)$$

Example 1:

The joint density of X and Y is given as

$$f(x,y)=\begin{cases}\dfrac{1}{\theta^2} & , \quad 0<y<\theta \text{ and } 0<x<\theta, \\ 0 & , \quad else\end{cases}$$

Note: X and Y are independent $Uniform(0,\ \theta)$ random variables.

a) What is the probability density of $U=aY+b$?

$F_U(u) = P(U < u) = P(aY+b < u) = P\left(Y < \frac{u-b}{a}\right)$

Case 1

$u > 0$

b) What is the probability density of $U = -a\ln(Y),\ a > 0$?

<u>Example 1:</u> (suppose $\theta = 1$)

The joint density of X and Y is given as

$$f(x,y) = \begin{cases} 1\ , & 0 < y < 1 \text{ and } 0 < x < 1, \\ 0\ , & else \end{cases}$$

e) What is the probability density of $U = -\ln(XY)$?

d) What is the probability density of $U = X + Y$?